

PT LAPI GANESHA UTAMA

In technical partnership with the Institute of Technology Bandung

TECHNICAL CERTIFICATION DOCUMENT

IECEX/ATEX Certification Document

Gas Leak Detector (GLD) V2

Prepared in direct reference to the IECEX/ATEX Certification Information Requirements issued by the certification body (ExCB) — covering Section 2, Technical Documentation (Items 1–6: product description; name, model, and specification list; functional description and technical parameters; product photographs; intended use and installation environment; and design and manufacturing information) and Section 3, Sample Information. Section 1, Basic Information (Application and Organization), remains to be completed in a subsequent revision.

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Technical partner	Institute of Technology Bandung
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Revision History

REVISION	DATE	DESCRIPTION
0.1	11 September 2026	Initial issue — Section 2, Items 1–6 (product description; name, model, and specification list; functional and technical parameters; product photographs; intended use and installation environment; design and manufacturing information) and Section 3 (sample information).

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About This Document

This document is a working technical file prepared to satisfy Section “2. Technical Documentation” and Section “3. Sample Information” of the official IECEx/ATEX Certification Information Requirements checklist issued by the certification body (ExCB). The table of contents follows the complete structure of the original checklist (Sections 1–3); Section 1 (Basic Information) remains pending and will follow in a subsequent revision.

Scope of this document. This is a compilation of design evidence and an honest readiness assessment against the ExCB checklist, prepared in support of a future submission — it is not itself a certificate and does not constitute self-certification. The final enclosure/casing (material selection, manufacture, gasket, and cable entry) is being developed by an external mechanical/casing partner and is not yet in the authors' possession; items that depend on that design (material datasheets, manufacturing process description, explosion-protection calculations) are reported as not yet available for that reason, not because the work has stalled. The product is still at from-scratch development/prototype stage — no finalized, serialized units exist yet, which is why a formal sample register (Section 3.1) is not yet available either.

Original excerpt, Section 2 & 3 — source: IECEx ATEX Certification Information Requirements (ExCB):

“1) Detailed product description; 2) Product name, model, and specification list; 3) Complete and clear functional description and technical parameters (electrical parameters, mechanical parameters, etc.); 4) Clear product photos (overall and key components); 5) Description of intended use and installation environment (e.g., gas group IIC/IIB/IIA if applicable, temperature group T1–T6, ambient temperature range, area classification 0/1/2 or 20/21/22); 6) Design and manufacturing information [a–i: drawings, BOM, material datasheets, manufacturing process, explosion-protection calculations, temperature group calculation, usage and installation instructions, nameplate information, Ex component certificates]. 3. Sample Information: 1) model, serial number, and status of the sample; 2) necessary test fixtures or auxiliary equipment.”

1. Basic Information (Application and Organization)

Covers the application form, business license/company registration, organizational chart and contact information, manufacturing facility address and profile, and (where applicable) ISO 9001 certification.

Not yet prepared in this revision. To be completed in a subsequent revision.

2. Technical Documentation

Six items per the original checklist (1–6, with Item 6 comprising sub-items a–i). This revision addresses the complete section: Items 1–6.

2.1 • Detailed Product Description

The Gas Leak Detector (GLD) is an IoT-based, multi-sensor gas leak detection device designed for the early detection of flammable and process gases in oil & gas refinery environments (process units, tank farms, pipe racks, and storage/loading-unloading areas). This document addresses the GLD unit itself as the subject of the current IECEx/ATEX certification.

Functionally, the GLD integrates eight channels of metal-oxide semiconductor gas sensors (MQ series), an edge-AI microcontroller/processor (ESP32-S3), and a LoRa radio module (star-topology wireless transmission) within a single fixed-point unit installed at locations with gas-leak risk. An on-device AI gas-classification model runs directly on the unit so that detection decisions do not depend on a continuous connection to a central server. When gas concentration exceeds a defined threshold, the unit triggers a local alarm (an integrated visual/audible alarm module) and simultaneously transmits an alarm notification over the LoRa network to the operator dashboard.

The enclosure is designed for hazardous-area deployment at refinery sites, using metal materials (aluminum alloy and stainless steel — no plastic or PVC) and mounted via an L-bracket to existing structures without drilling or welding. Important: this design-intent statement does not constitute a claim that the enclosure has passed testing or has been Ex-certified — the explosion-protection scheme, gas group, temperature class, and target installation zone will be addressed in Section 2.5 (to follow).

The current production power configuration is continuous 24 VDC, supplied via an AC/DC adapter connected to the site electrical supply. A portable battery power path (Li-ion 18650) remains under development (R&D) and has not become a deployed production configuration.

2.2 · Product Name, Model, and Specification List

Product name	Gas Leak Detector (GLD) — Node Sensor
Model / version	GLD V2 (Version 2)
Manufacturer	PT LAPI Ganesha Utama
Technical development partner	Institute of Technology Bandung — IoT Laboratory & Physics Laboratory
End client / program owner	PT Pertamina Patra Niaga (initial deployment site: Refinery Unit IV, Cilacap)
Primary function	Acquisition of 8-channel gas sensor data and LoRa transmission
Microcontroller	ESP32-S3-WROOM-1U-N16R8
LoRa radio module	E22-900MM22S
Dimensions (L×W×H)	200 × 90 × 290 mm
Enclosure material	Aluminum alloy + stainless steel

Source: official product technical datasheet (Institute of Technology Bandung, Revision 4.0), cross-referenced with internal technical specification documentation and EMC parameter measurement data.

2.3 · Functional Description and Technical Parameters (Electrical, Mechanical, etc.)

Functional workflow (normal operating mode): sense → process → transmit. Each of the eight gas sensors continuously samples ambient conditions → data is normalized and classified by the on-device AI model (ESP32-S3) → the result is transmitted over the LoRa network (star-topology transmission) at a configurable interval (default 10 seconds), or immediately (event-driven) when an alarm condition is detected. Gas alarms are triggered through two parallel channels: a local visual/audible alarm module on the unit itself, and a push notification transmitted over the LoRa network to the dashboard — the alarm-push pathway has been

successfully tested on a campus mesh network (field validation at a production refinery installation is still pending).

2.3.a • Electrical Parameters — Node Sensor (GLD)

PARAMETER	SPECIFICATION	STATUS	REMARKS
Main power input	24 VDC	Final — production config.	Continuous power supply from a site AC/DC adapter.
AC/DC adapter input	220 VAC, 50 Hz	Final	Compliant with the Indonesian national electrical grid standard (PLN).
AC/DC adapter output	24 VDC	Final	
Internal operating voltage	5 VDC & 3.3 VDC	Final	5 VDC for the MQ sensor/heater circuitry; 3.3 VDC for ESP32-S3 logic. Per-rail current not yet documented separately.
Maximum input current	≈0.33 A @ 24 VDC	Calculated	Calculated from measured maximum power consumption (7.995 W) divided by 24 VDC.
Maximum power consumption	7.995 W @ 24 VDC	Measured — production config.	Applies to the continuous-power configuration. The battery (R&D) configuration is recorded separately at 5.75 W — a different operating mode, not a data conflict.
Electrical protection (fuse, reverse polarity, overvoltage, overcurrent)	2× resettable PPTC fuse (F1/F2); TVS/ESD suppression diodes (D1, D4/D5/D11, D6); Schottky diodes (D7/D13, D8/D9, D12)	Partially available — component-level evidence	Real components exist per the EasyEDA/JLPCB bill of materials (Section 2.6.b): 2× Littelfuse MINISMD260F/16 resettable fuses (F1, F2, overcurrent), a Ruilong SMBJ33A TVS diode (D1) and 3× UMW LESD5D5.0CT1G / 1× DOWO SM712 ESD-protection arrays (D4, D5, D11, D6), and multiple Schottky diodes — MDD SS54 (D7, D13), MDD SS14 (D8, D9), GOOD-ARK SK36 (D12). This is genuine evidence that overcurrent and transient/ESD protection circuitry exists on the board; which specific rail each component protects and its exact circuit role (e.g., reverse-polarity blocking vs. flyback) has not yet been cross-checked against the schematic net list, and no consolidated protection-scheme write-up has been produced for ExCB review.

2.3.b · Sensor and Communication Parameters — Node Sensor (GLD)

PARAMETER	SPECIFICATION	STATUS	REMARKS
Gas sensors	MQ-2, MQ-3B, MQ-4, MQ-5, MQ-6, MQ-7B, MQ-8, MQ-135 (8 channels)	Final	Metal-oxide semiconductor sensors; sensing element directly exposed to ambient air.
AI gas-classification model	On-device classifier — 3 classes: Clean Air, LPG, H ₂	Final	Runs locally on the ESP32-S3 (Running/Inference mode); outputs a class label and a confidence value. Does not yet cover CO ₂ , Benzene, CO, or H ₂ S.
Environmental sensor (temperature/humidity)	SHT40-AD1B-R2 (I2C)	Final	Auxiliary temperature/humidity input included in the primary hardware design; values are available for status/telemetry.
Analog-to-digital converter	ADS1256IDBR	Final	24-bit multi-channel analog acquisition for the 8 sensor channels; 30,000 SPS, firmware SPI clock 1.92 MHz.
Processing unit	ESP32-S3-WROOM-1U-N16R8	Final	Certified under FCC (2AC7Z-ESPS3WROOM1U), TELEC, and CE (per Espressif data) — RF/EMC certifications, not an “Ex component” certification.
Communication module	LoRa, E22-900MM22S module	Final	Certified under CE, FCC, and RoHS (per Ebyte data) — RF/EMC certifications, not an “Ex component” certification.
Operating frequency	920 MHz	Final	Star-topology transmission; within the regional 920–923 MHz ISM band (Indonesia).
Transmit power (firmware configuration)	17 dBm	Final	The radio module supports up to 22 dBm — 17 dBm is an operational configuration, not the module's maximum limit.
Bandwidth / spreading factor / coding rate	125 kHz / SF7 / CR 4/5	Final	Source: EMC parameter table and official product technical datasheet.
Antenna	External, omnidirectional, SMA male connector, 3 dBi gain	Final	On some units, the 2.4 GHz Wi-Fi antenna remains inside the enclosure and must be relocated externally.
Data transmission interval	Configurable, default 10 seconds	Final	Alarm events are transmitted immediately, independent of the periodic interval.
Other interfaces	SPI, LoRa	Final	External ports/connectors: USB, sensor, power, fan, antenna, alarm

			buzzer.
RS-485 / Modbus interface	Read-only Modbus RTU slave, 9600 bit/s 8N1, Unit ID 1 (THVD1410DR transceiver)	Final	Provides 8 read-only registers (device status word, classification result, confidence, battery voltage, power source, external power, LoRa transmission counter, node ID); not used for product control.

2.3.c · Mechanical Parameters — Node Sensor (GLD)

PARAMETER	SPECIFICATION	STATUS	REMARKS
Enclosure material	Aluminum alloy + stainless steel	Partially available — specific grade pending	Per project confirmation, the production enclosure is sourced from a commercially available CE/ATEX-marketed explosion-proof gas-detector housing product line (referenced supplier listing: Alibaba.com, “CE ATEX Explosion Proof H2 Sensor”), consistent with the cast-metal housing, threaded “Ex”-marked cable entry, and sensor mesh cover shown in the product photography (Section 2.4). This is corroborated by an internal case CAD drawing (“GLD ATEX CASE v3,” dated 8 September 2026) specifying a cylindrical sensor-case body (Ø102 mm outer housing ring, Ø90/Ø80 mm internal bores) that incorporates a stainless-steel filter mesh disc, a small DC cooling fan, and a transparent viewing window. The specific alloy/grade of the aluminum body and the supplier listing's own certification claims have not been independently verified — the listing's specification text could not be retrieved for cross-check, so this should be treated as project-confirmed sourcing context, not a verified datasheet citation. PVC is not used in any housing or bracket component.

Dimensions (L × W × H)	200 × 90 × 290 mm	Final	Consistent between the technical specification documentation and the EMC parameter table.
Total weight	—	Pending confirmation	Not yet weighed/documented.
Mounting method	L-bracket, following the design already installed at the refinery	Final	Mounted to existing structures without drilling or welding.
Ingress protection (IP rating)	—	Pending confirmation	Not yet tested/determined.
Cable entry (gland)	—	Pending confirmation	Cable gland specification not yet determined.
Antenna mounting	External, SMA male connector	Final	
Operating temperature	—	Pending confirmation	Ambient operating temperature range not yet determined — key parameter for temperature class (T1–T6) determination in Section 2.5.
Operating humidity	—	Pending confirmation	

Source: internal technical specification documentation, Sections 1.1–1.3 (Node Sensor), cross-referenced with the official product technical datasheet (Revision 4.0) and component certification verification (ESP32-S3-WROOM-1U, E22-900MM22S).

Rows marked “Pending confirmation” above do not reflect a documentation oversight — this is an honest status indicator. These fields are intentionally left blank because no official data yet exists (not yet measured, tested, or decided). They must not be filled with estimates in future revisions without a clear supporting data source.

2.4 · Product Photographs — Overall and Key Components

The seven photographs below were taken directly from the Node Sensor (GLD) V2 prototype unit (no capture-date metadata is available in the source files). The first three photographs show the fully assembled unit from different angles; the remaining four show key components in close-up.



Complete unit — front view

The fully assembled Node Sensor (GLD): the main enclosure (blue) with sensor mesh cover, external LoRa antenna, and a visual alarm module (cylindrical housing, red lens) mounted on the right side.



Complete unit — alternate angle

An alternate view of the assembled unit, showing the position of the antenna, alarm module, and the cable-gland side of the enclosure.



Complete unit — front view (without antenna and alarm module)

The main enclosure with the sensor mesh cover installed, shown in a flat front view illustrating the enclosure shape and its four mounting bolt holes.



Key component — PCB inside enclosure (before sensor modules)

The main PCB installed inside the enclosure prior to gas sensor module installation. The ESP32-S3-WROOM-1U microcontroller, LoRa radio module, and power circuitry are visible.



Key component — PCB with eight gas sensor modules

The same PCB with eight channels of MQ-series gas sensors (MQ-2, MQ-3B, MQ-4, MQ-5, MQ-6, MQ-7B, MQ-8, MQ-135) installed around the central microcontroller.



Key component — visual alarm module (close-up)

The visual alarm (beacon/strobe) module, constructed of metal with a red lens, connected to the main enclosure via a dedicated cable gland.



Key component — sensor mesh cover (close-up)

The metal mesh cover that protects the gas sensing elements while serving as a diffusion path for gas to reach the sensors beneath.

Photographic completeness — status as-is. The seven photographs above cover the complete unit and its key components (PCB, sensor modules, alarm module, mesh cover), substantively satisfying checklist Item 2.4. Still outstanding: (a) formally labeled photographs of each face (front/back/left/right/top/bottom) with a scale reference, as is customary in ExCB submission packages; (b) separate photographs of individual components such as the battery, gaskets/seals, terminals, and cable glands.

2.5 • Description of Intended Use and Installation Environment (Gas Group, Temperature Class, Area Classification)

The GLD is intended for continuous, fixed-point monitoring of flammable and toxic gas leaks in oil & gas refinery hazardous areas — process units, tank farms, pipe racks, and loading/unloading areas. The specific Ex

classification parameters below reflect the project engineering team's current recommendation, developed from the intended deployment envelope and the gases the device is designed to detect; none of these parameters has yet been confirmed with the certification body (ExCB) or a notified body, and none should be treated as final.

PARAMETER	RECOMMENDED VALUE	STATUS	RATIONALE / REMARKS
Gas group	IIC	Team recommendation — pending ExCB confirmation	Hydrogen (H ₂) is one of the three gases the classifier is designed to detect (Section 2.3.b); IIC is required for hydrogen and inherently covers IIB and IIA.
Temperature class	T4 (≤135°C)	Team recommendation — pending verification	Not yet substantiated by measurement. MQ-series metal-oxide sensors rely on an internal heating element as their normal operating principle; a hot-spot measurement specific to the sensor models and enclosure configuration used here has not yet been performed (see 2.6.f).
Ambient temperature range	—	Pending confirmation	Not yet measured or specified.
Area classification (zone)	Zone 1	Team recommendation — pending ExCB confirmation	Assessed as sufficient for the general refinery deployment envelope, including areas near storage. Zone 0 would only apply if the detector were installed directly inside a tank vapor space.

These are engineering recommendations, not a certification decision. An independent readiness assessment of the same checklist item, prepared separately from the underlying firmware/hardware repository, reached the same conclusion — gas group, temperature class, ambient range, and zone are not yet formally established. Final classification requires explicit agreement with the ExCB.

2.6 • Design and Manufacturing Information

Nine sub-items (a–i) per the original checklist. Status is reported item by item below; most sub-items are not yet available — this is reported plainly rather than implied to be complete.

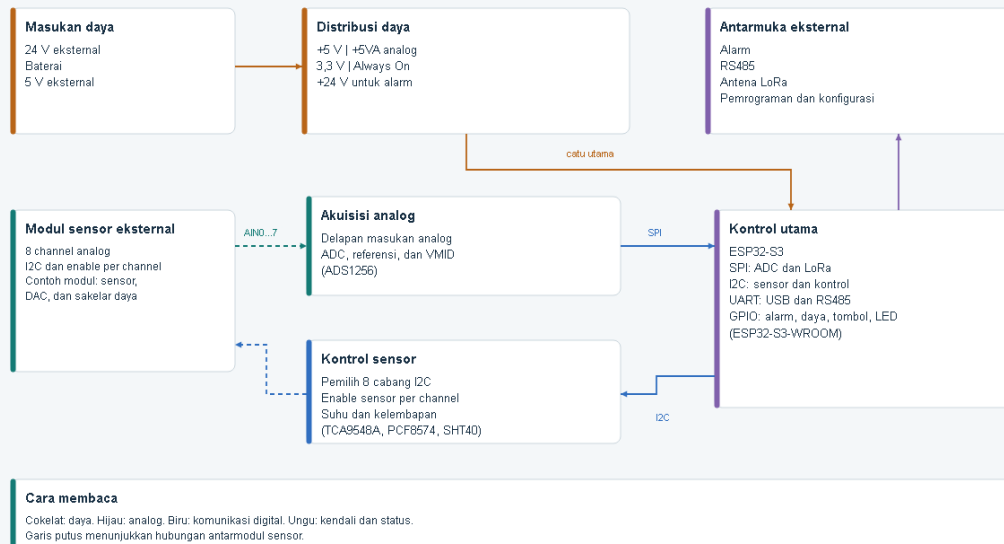
2.6.a • Complete Drawings (Assembly, Component, Electrical Schematic, PCB Layout, Enclosure Structure, Junction Box, Terminal, Grounding)

An electrical schematic capture and a corresponding PCB layout exist for the GLD V2 main board as native EasyEDA/JLPCB source design files (not just a derived summary). From the schematic's traced net list, a supporting block-diagram set (9 sheets, functional/block level, 204 components mapped with documented pin-to-net traceability) has been produced and is reproduced in full below. Field labels in the source diagrams are in Indonesian; English captions are provided under each sheet.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

01 Arsitektur keseluruhan

Catu daya, akuisisi sensor, kendali, dan komunikasi dalam satu perangkat.



01 / 9

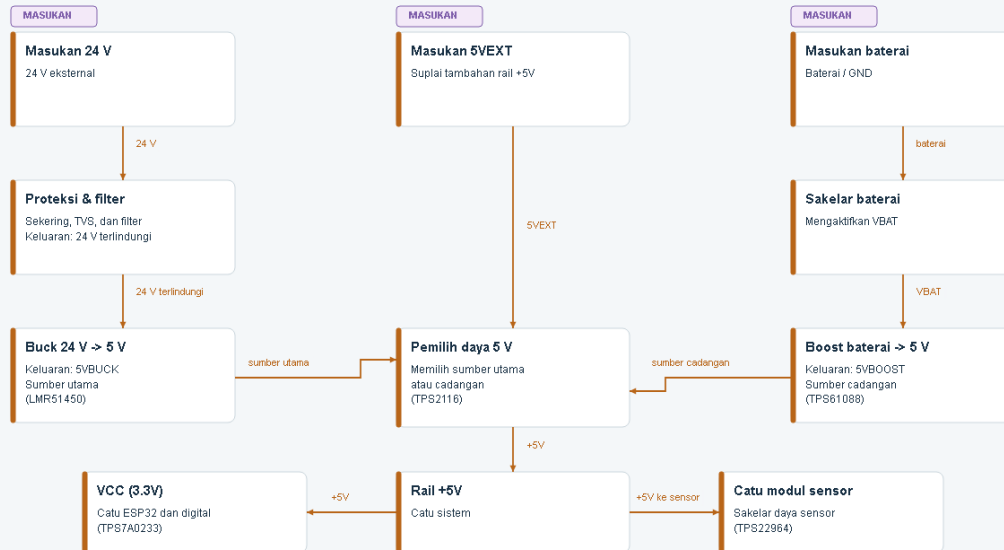
Sheet 1 of 9 — Overall architecture

Power input, power distribution, analog acquisition, sensor control, main control (ESP32-S3), and external interfaces (alarm, RS-485, LoRa antenna, programming/configuration).

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

02 Distribusi daya utama

Tiga masukan daya, pemilihan sumber 5 V, rail analog, dan catu sensor.



24 V terlindungi: keluaran setelah sekering, TVS, dan filter.

02 / 9

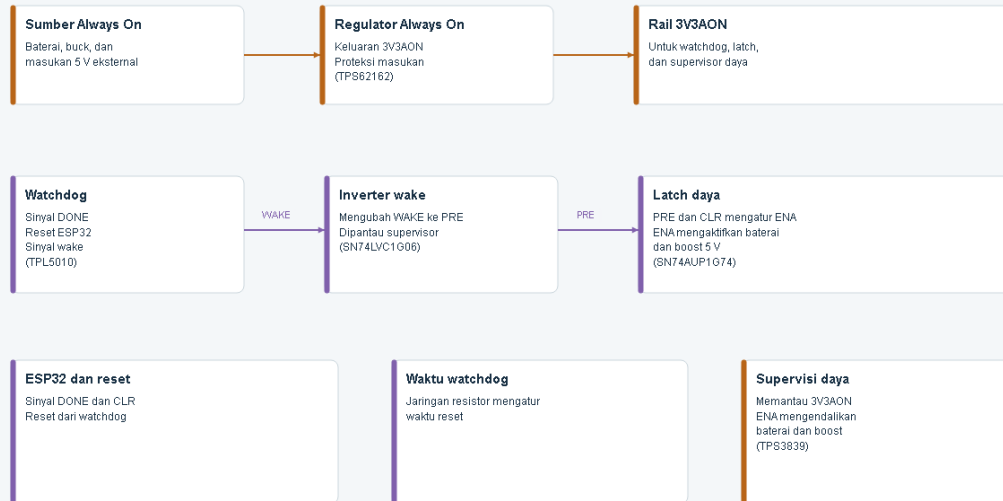
Sheet 2 of 9 — Main power

Main power input rails and distribution: +5 V / +5VA analog, 3.3 V, and +24 V for the alarm circuit.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

03 Always On, watchdog, dan latch

Always On (AON) adalah rail daya yang tetap aktif untuk menjaga pengawasan dan pengendalian daya.



03 / 9

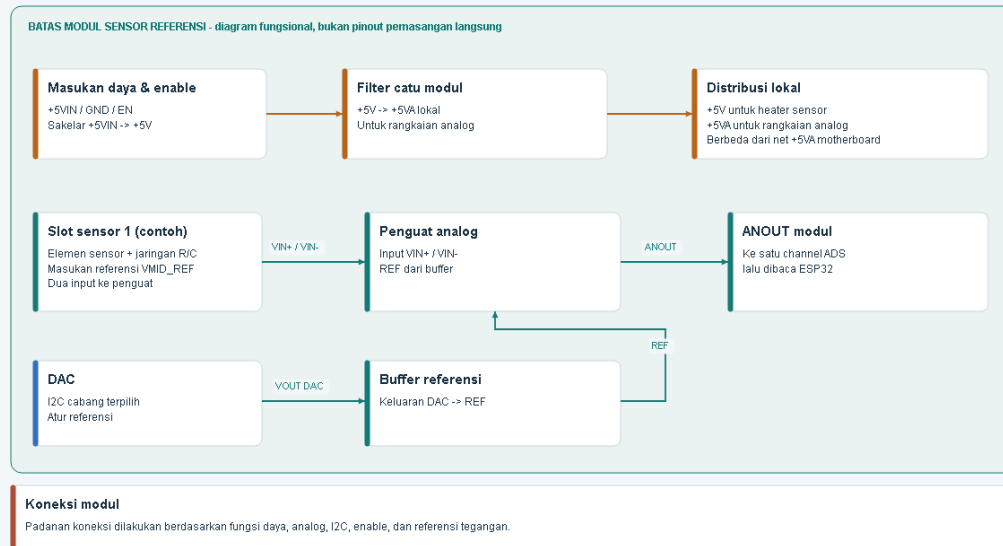
Sheet 3 of 9 — Always-on power

The always-on power domain that remains active independent of the main control state.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

04 Modul sensor: jalur fungsional

Setiap modul menerima daya, I2C, enable, dan menyediakan satu keluaran analog.



04 / 9

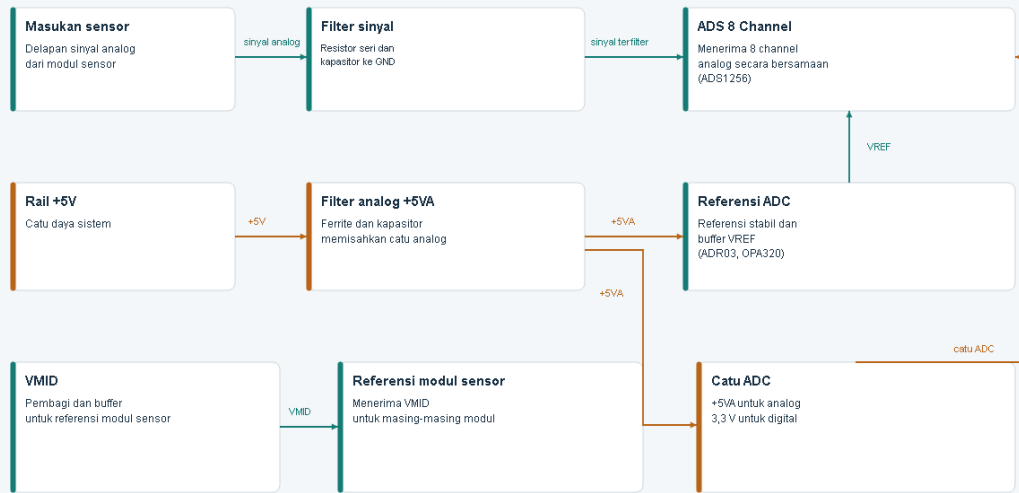
Sheet 4 of 9 — Sensor module

The external gas sensor module: 8 analog channels with per-channel I2C enable, including the sensor, DAC, and power-switch sub-blocks.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

05 Akuisisi analog dan tegangan referensi

Jalur analog memiliki filter catu terpisah, referensi tegangan, VMID, dan ADS 8 Channel.



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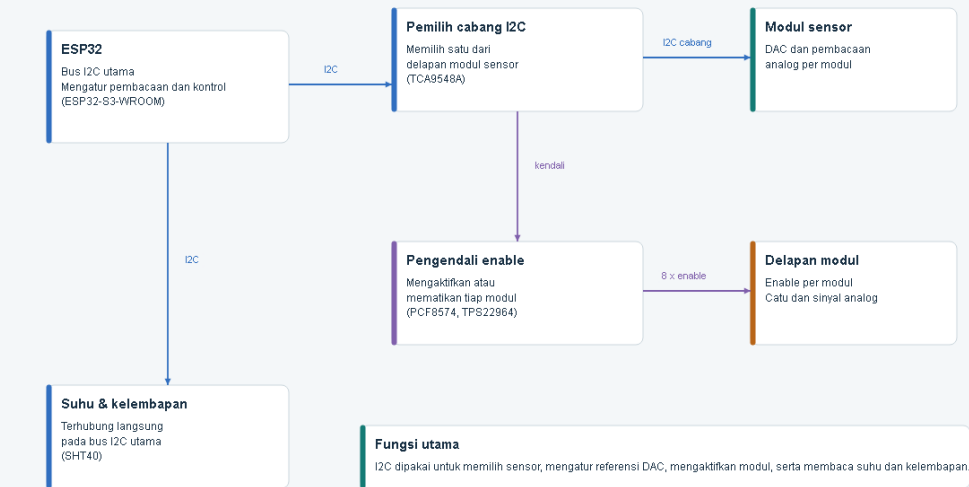
Sheet 5 of 9 — Analog acquisition

The 8-channel analog acquisition front end: analog inputs, ADC (ADS1256), reference, and VMID biasing.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

06 I2C dan kendali sensor

I2C memilih cabang sensor, mengatur DAC, dan mengendalikan daya tiap modul.



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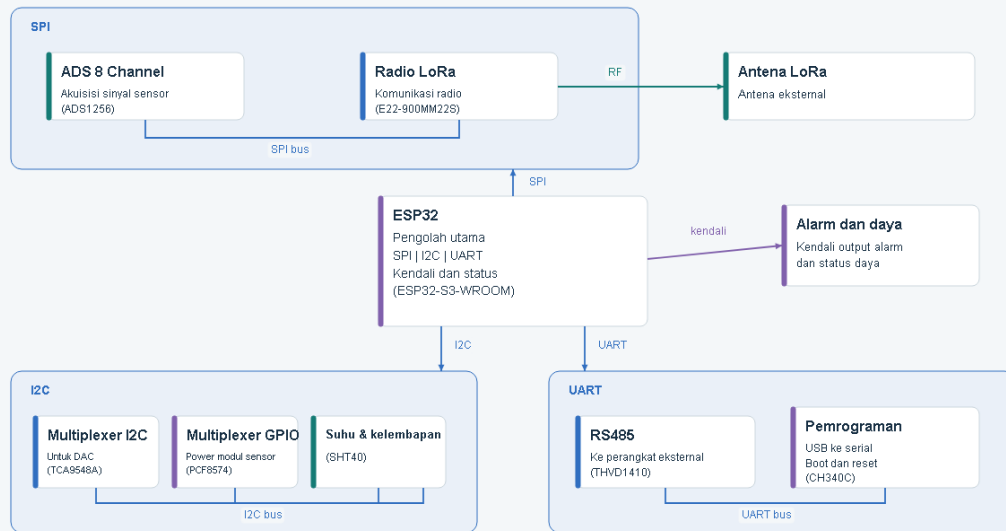
Sheet 6 of 9 — I2C & sensor control

I2C channel selection (TCA9548A), per-channel sensor enable (PCF8574), and the temperature/humidity sensor (SHT40).

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

07 Komunikasi dan koneksi ESP32

Koneksi dikelompokkan menurut bus SPI, I2C, dan UART, kendali perangkat dipisahkan dari komunikasi.



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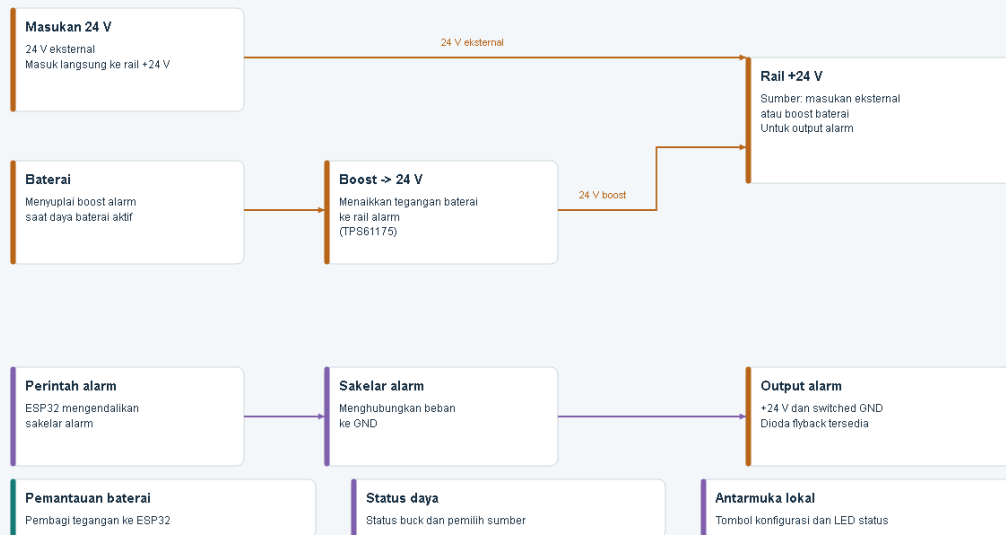
Sheet 7 of 9 — ESP32 connections

Main controller (ESP32-S3) interface map: SPI to the ADC, I2C to sensor/control, UART for USB and RS-485, and GPIO for alarm, power, button, and LED.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

08 Alarm, baterai, dan status daya

Rail +24 V alarm dapat berasal dari masukan 24 V eksternal atau boost baterai.



08 / 9

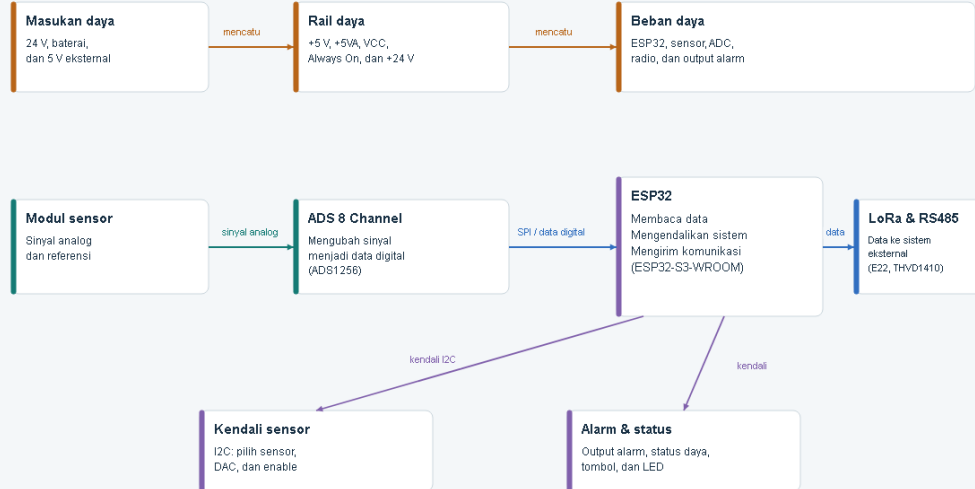
Sheet 8 of 9 — Alarm & status

Alarm output and status-indication circuitry driven from the main controller.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

09 Alur energi, data, dan kendali

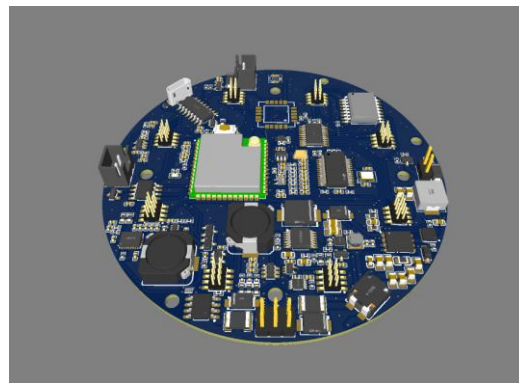
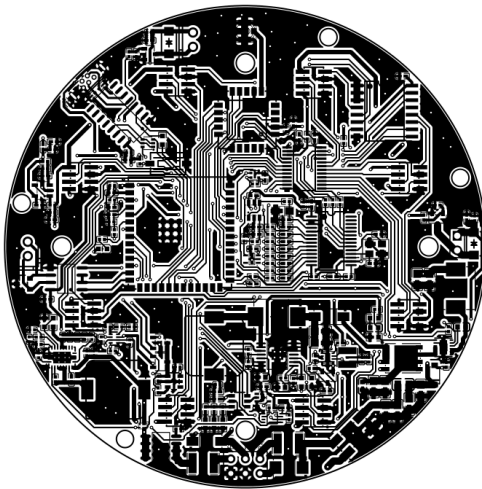
Daya mencatu perangkat; data sensor mengalir ke ESP32; ESP32 mengendalikan komunikasi dan keluaran.



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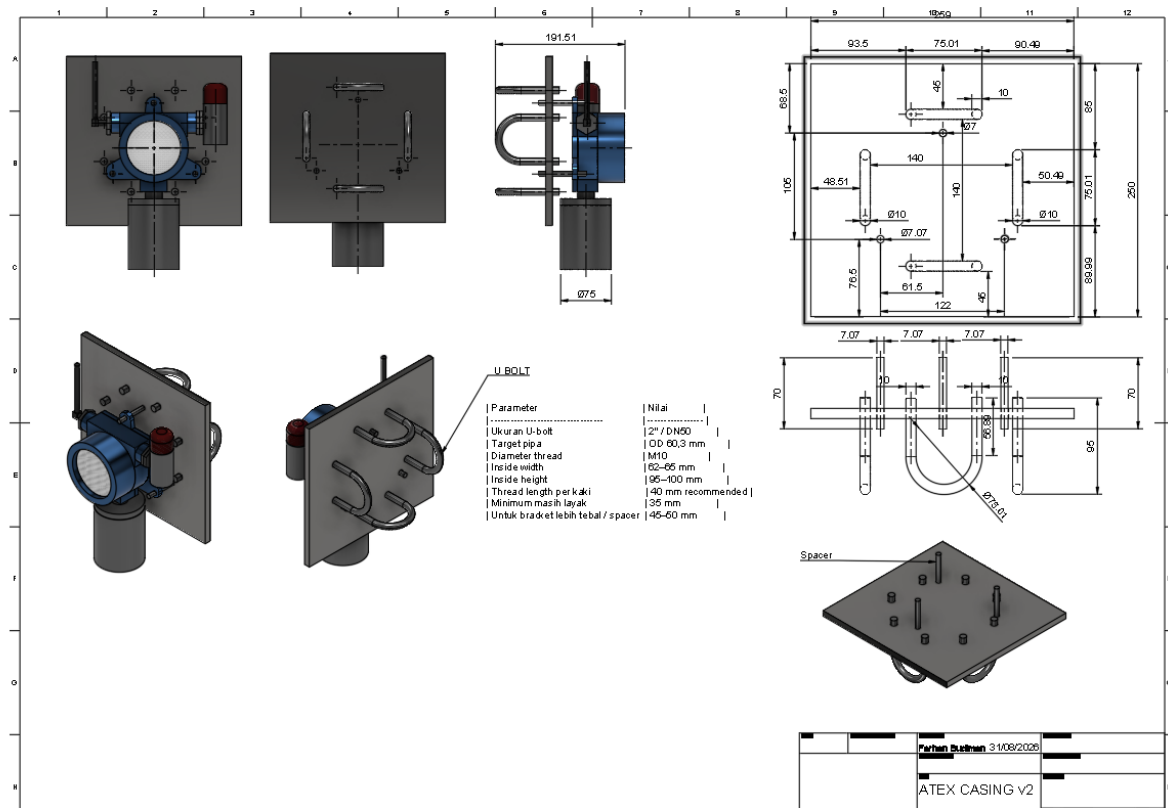
Sheet 9 of 9 — System flow

Overall signal and data flow across the board, summarizing how the preceding eight sheets connect end to end.

**Main board PCB layout (top copper) and 3D populated render**

Both exported directly from the EasyEDA/JLCPCB source project (production-intent board, circular outline with six mounting holes). The 3D render shows the ESP32-S3-WROOM module, micro-USB connector, power inductors, and 8-channel I2C header block in their real placed positions; it is an illustrative render, not a dimensioned drawing.

A dimensioned mechanical drawing sheet also exists for the enclosure's external envelope and its mounting hardware, drafted from a solid CAD model (STEP format, millimeter units) with a formal title block, orthographic and isometric views, and a parts table — reproduced below.



Enclosure envelope & mounting bracket — dimensioned drawing sheet

Orthographic and isometric views with real dimensions (enclosure neck $\varnothing 75$ mm, overall probe height 191.51 mm, mounting plate 250x250 mm with toleranced hole pattern), a U-bolt parameter table (2"/DN50, M10 thread), and a title block (drafted 31 Aug 2026). Source: dimensioned CAD drawing derived from a STEP solid model of the enclosure and bracket assembly.

DRAWING TYPE	STATUS	REMARKS
Electrical schematic (component-level) / block diagram	Partially available	Schematic capture and a derived 9-sheet block-diagram set exist with traceability evidence (pin-to-net mapping). Not yet issued in a released, revision-controlled drawing format with a formal drawing number.
PCB layout	Partially available	Native EasyEDA/JLPCB layout export (routed copper) and a 3D solid model exist for the same board revision. Not yet issued as a dimensioned, toleranced, released drawing with a formal drawing number.
Assembly drawing	Partially available	A dimensioned drawing sheet exists for the enclosure/bracket mounting assembly (title block, orthographic + isometric views, parts table). It covers the external envelope and mounting hardware, not the internal PCB/component assembly sequence.
Component drawing	Not yet available	
Enclosure structure drawing (gap, length, volume)	Partially available	External envelope is dimensioned (from a real STEP solid model, millimeter units) in the

		drawing above. The Ex-d-specific flame-path parameters — joint gap, length, and free internal volume — are not called out; that dimensioning has to be added deliberately once a protection concept is confirmed (Section 2.5), not derived automatically from the CAD export.
Junction box / terminal / grounding connection drawings	Not yet available	

Real dimensioned CAD drawings now exist for the enclosure's external envelope and mounting hardware, and native PCB layout/schematic source files exist for the electronics. What is still missing for a certifiable drawing package: internal component/assembly drawings, junction box and grounding-connection drawings, and — specific to explosion protection — flame-path gap/length/volume dimensioning, which requires a confirmed protection concept before it can be drawn.

2.6.b · Bill of Materials (BOM) for Explosion-Safety-Relevant Components

A complete, itemized electronic-component BOM for both the main board and the external sensor board now exists, exported directly from the EasyEDA/JLCPCB source project (manufacturer, manufacturer part number, and LCSC supplier part number for each line item; 89 line items / 214 placed components on the main board, 14 line items / 24 placed components on the sensor board — full tables below). This is real, traceable sourcing data and materially improves on the previous status.

It does not, however, satisfy this checklist item as written. The checklist asks specifically for the explosion-safety-relevant BOM — enclosure, gaskets, terminals, cable entry devices, switches, light sources, battery, potting compound, and plastic parts, each with material grade and Ex/UL/CCC certification status. None of those mechanical/safety items appear in an electronic CAD BOM: the enclosure is present only as a placeholder mechanical symbol (designator U50, no manufacturer or dimensional data attached), and the gas sensor itself (MQ2, designator I1) has no manufacturer or supplier part number recorded — it is sourced outside the LCSC/JLCPCB supply chain and its Ex status is unverified. Comparative research on Ex-rated enclosure products from other manufacturers exists internally as a reference for target specifications only — it describes third-party products, not this product's actual components, and is not included here.

DESIGNATOR	VALUE / PART	QTY	MANUFACTURER	MANUFACTURER PART NO.	LCSC #
C1	1uF	1	SAMSUNG	CL21B105KBFNNNE	C28323
C2,C3	100nF	2	YAGEO	CC0805KRX7R9BB104	C49678
C4,C59,C60,C10	10uF	4	muRata	GRM21BR61H106KE43L	C440198
C72,C73	33uF	2	TDK	C2012X5R1A336MTJ00E	C342635
C7	47nF	1	SAMSUNG	CL10B473KB8NNNC	C1622
C8	22nF	1	SAMSUNG	CL10B223KB8NNNC	C21122
C70	33pF	1	SAMSUNG	CL10C330JB8NNNC	C1663
C61,C69	100nF	2	YAGEO	CC0603KRX7R9BB104	C14663
C87,C12,C13,C14,C41,C42,C43,C44,C46,C47,C48,C49,C50,C51,C52,C54,C55,C	100nF	38	SAMSUNG	CL05B104KO5NNNC	C1525

58,C65,C66,C15,C6, C19,C20,C21,C22,C 23,C24,C25,C26,C2 7,C28,C29,C30,C31, C84,C78,C81					
CN3,CN1	XH-2A-black	2	BOOMELE	XH-2A 黑色	C24193
D1	SMBJ33A	1	RUILON	SMBJ33A	C10219
D2	LBZT52C12T1G	1	LRC	LBZT52C12T1G	C12747
D3,D10	BAT54C	2	CJ	BAT54C	C2135
D4,D5,D11	LES5D5.0CT1G_C5199850	3	UMW	LES5D5.0CT1G	C5199850
D6	SM712	1	DOWO	SM712	C293966
D7,D13	SS54_C22452	2	MDD	SS54	C22452
D12	SK36_C266330	1	GOOD-ARK	SK36	C266330
F1,F2	MINISMDC260F/16	2	Littelfuse	MINISMDC260F/16-2	C16490
G\$1	UFL	1	—	—	—
J1	VIN	1	—	—	C65114
L1	300Ω	1	TDK	ACM7060-301-2PL-TL01	C76580
L2	2.2uH	1	SXN	SMDRH105R-2R2NT	C10164
L3	2.2uH	1	APV	ANR3015T2R2M	C6364793
L4	100Ω	1	TDK	MPZ2012S101AT000	C15957
L5	10uH	1	SXN	SMDRH105R-100MT	C9185
LED1	19-217/BHC-ZL1M2RY/3T	1	EVERLIGHT	19-217/BHC-ZL1M2RY/3T	C72041
Q1	UMH3NTN	1	ROHM	UMH3NTN	C123008
Q2	AO4407_C5155210	1	JSMSEMI	AO4407	C5155210
Q3	SI7465DP-T1-GE3	1	VISHAY	SI7465DP-T1-GE3	C499482
R3	187kΩ	1	UNI-ROYAL	0603WAF1873T5E	C22896
R6	82kΩ	1	UNI-ROYAL	0603WAF8202T5E	C23254
R86	19.1kΩ	1	UNI-ROYAL	0603WAF1912T5E	C22897
R87	1kΩ	1	UNI-ROYAL	0603WAF1001T5E	C21190
R89,R36	56kΩ	2	UNI-ROYAL	0603WAF5602T5E	C23206
R90	189k	1	YAGEO	RT0603BRD07189KL	C861174
R92	33kΩ	1	UNI-ROYAL	0603WAF3302T5E	C4216
R96	470Ω	1	UNI-ROYAL	0603WAF4700T5E	C23179
SW1	SKQYAAE010	1	ALPSALPINE	SKQYAAE010	C115352
U1	TPS61175PWP	1	TI	TPS61175PWP	C2879383
U36	LMR51450SDRRR	1	TI	LMR51450SDRRR	C6890985
U40	4.7uH	1	ABC	GSSM06304R7M2AU	C2928306
U41	TPS61088RHLR	1	TI	TPS61088RHLR	C87357
U42	TPS7A0233DBVR	1	TI	TPS7A0233DBVR	C5142805
U43	TPS62162DSG	1	TI	TPS62162DSGR	C40256
U46	TPL5010	1	TI	TPL5010DDCR	C473912
U47	THVD1410DR	1	TI	THVD1410DR	C2671345
U48	CH340C	1	WCH	CH340C	C84681
U50	casingAlGasLeak	1	—	—	—
U51	SN74AUP1G74DCUR	1	TI	SN74AUP1G74DCUR	C140267
USB1	U-D-M5DD-W-1	1	韩国韩荣	U-D-M5DD-W-1	C145795
C11,C53,C32,C33,C 34,C35,C36,C37,C1 6,C17,C18,C38,C85, C79,C80	10uF	15	SAMSUNG	CL05A106MQ5NUNC	C15525

C45,C57	20pF	2	FH	0402CG200J500NT	C1554
C56	22uF	1	Kyocera AVX	TAJA226M010RNJ	C7183
R8,R9,R10,R74	22Ω	4	UNI-ROYAL	0402WGF220JTCE	C25092
R11,R65,R66,R67,R68,R69,R70,R71,R72	49.9Ω	9	FH	RC-02K49R9FT	C140147
R73,R75,R77,R2,R15,R16,R17,R18,R19,R20,R21,R22,R7,R28,R29	10kΩ	15	UNI-ROYAL	0402WGF1002TCE	C25744
R76,R5,R31	1kΩ	3	UNI-ROYAL	0402WGF1001TCE	C11702
U2,U3	OPA320AIDBVR	2	TI	OPA320AIDBVR	C92494
U33	TCA9548APWR	1	TI	TCA9548APWR	C130026
U35	ADS1256IDBR	1	TI	ADS1256IDBR	C28186
U38	ADR03AKSZ-REEL7	1	ADI	ADR03AKSZ-REEL7	C206683
U39	8MHz	1	JYJE	3TJ480000NYLBC	C2924618
U44	SHT40-AD1B-R2	1	Sensirion	SHT40-AD1B-R2	C2909890
U45	E22-900MM22S	1	EBYTE	E22-900MM22S	C5333985
U49	ESP32-S3-WROOM-1U-N16R8	1	ESPRESSIF	ESP32-S3-WROOM-1U-N16R8	C3013946
U12	PCF8574T/TR	1	HGSEMI	PCF8574T/TR	C2987288
R13	16kΩ	1	UNI-ROYAL	0402WGF1602TCE	C25759
R14	4.99kΩ	1	YAGEO	RC0402FR-074K99L	C137965
U13	TPS2116DRLR	1	TI	TPS2116DRLR	C3235557
U14	TPS22964CYZPR	1	TI	TPS22964CYZPR	C431579
U15	SN74LVC1G06DCKR	1	TI	SN74LVC1G06DCKR	C7828
U16	TPS3839K33DBZR	1	TI	TPS3839K33DBZR	C96333
C9	22uF	1	TDK	C2012X5R1V226MT000E	C338080
R23,R24,R25,R1,R26,R27,R4,R30,R12,R40,R41,R43	100kΩ	12	UNI-ROYAL	0402WGF1003TCE	C25741
R32	200kΩ	1	UNI-ROYAL	0402WGF2003TCE	C25764
C39,C40,C62,C63,C64,C67,C68,C5	22uF	8	TAIYO YUDEN	JDK105CBJ226MV-F	C385994
C71	1uF	1	SAMSUNG	CL05A105KA5NQNC	C52923
R33,R37,R38,R39	0Ω	4	UNI-ROYAL	0402WGF0000TCE	C17168
C74	2.2uF	1	SAMSUNG	CL05A225MQ5NSNC	C12530
C75,C76	47nF	2	FH	0402B473K500NT	C82219
R34	255kΩ	1	YAGEO	AC0402DR-07255KL	C226702
R35	43kΩ	1	UNI-ROYAL	0603WAF4302T5E	C23172
C77	4.7nF	1	FH	0402B472K500NT	C1538
D8,D9	SS14	2	MDD	SS14	C2480
J2	ALARM	1	—	—	C124375
Q4,Q5	AO3400A	2	AOS	AO3400A	C20917
R94,R42	100Ω	2	UNI-ROYAL	0402WGF1000TCE	C25076
H2,H1,H3,H4,H5,H6,H7,H8	X1321WVS-2X04J-C40D53	8	XKB Connection	X1321WVS-2x04J-C40D53	C2881909
L6	60Ω	1	Sunlord	GZ1005D600TF	C23478

External sensor board (gas-sensing front end):

DESIGNATOR	VALUE / PART	QTY	MANUFACTURER	MANUFACTURER PART NO.	LCSC #
H1	X1321FVS-2X04-C43D48	1	XKB Connection	X1321FVS-2x04-C43D48	C2684732
R9	1kΩ	1	UNI-ROYAL	0402WGF1001TCE	C11702
U5	TPS22919DCKR	1	TI	TPS22919DCKR	C2149796
I1	MQ2	1	—	MQ2	—
C7,C8	10uF	2	SAMSUNG	CL05A106MQ5NUNC	C15525
L1	60Ω	1	Sunlord	GZ1005D600TF	C23478
R1	10MΩ	1	UNI-ROYAL	0402WGJ0106TCE	C26086
R2,R3	10kΩ	2	UNI-ROYAL	0402WGF1002TCE	C25744
R7,R8	20kΩ	2	UNI-ROYAL	0402WGF2002TCE	C25765
C1,C2,C3,C4,C5,C6	100nF	6	SAMSUNG	CL05B104KO5NNNC	C1525
R4,R5,R6	49.9Ω	3	FH	RC-02K49R9FT	C140147
U1	INA333AIDGKR	1	TI	INA333AIDGKR	C19450
U2	LM321DTR(XBLW)	1	XBLW	LM321DTR(XBLW)	C18723612
U3	MCP4725A0T-E/CH	1	MICROCHIP	MCP4725A0T-E/CH	C144198

Status: Partially available. Full electronic-component BOM with real manufacturer/supplier data now exists (source: EasyEDA/JLPCB export, 11 Sep 2026). The explosion-safety-relevant subset the checklist actually asks for — enclosure, gasket, cable entry device, battery, potting compound, and the gas sensor itself, with material grade and Ex/UL/CCC certification for each — remains not yet compiled. These are mechanical/safety parts, not electronic components: the enclosure itself is under design by an external mechanical/casing development partner and its bill of materials has not yet been provided to the authors.

2.6.c • Material Specification Sheets / Datasheets (Non-Metallic Materials)

Datasheets or supplier conformity declarations for non-metallic materials (enclosure components, seals, insulators, potting compounds) — covering heat/cold resistance, anti-aging, anti-static, flame retardancy, CTI value, and chemical resistance — have not yet been collected.

Status: Not yet available. Dependent on the enclosure design and material selection, which sits with the external casing development partner rather than the authors.

2.6.d • Manufacturing Process Description

A description of manufacturing processes relevant to explosion-protection safety (enclosure machining accuracy control, explosion-proof surface treatment, welding, potting, die-casting, bonding) has not yet been documented.

Status: Not yet available. This describes the casing partner's manufacturing process, not an internal electronics process — it will need to be obtained from that partner once their process is finalized.

2.6.e • Explosion-Protection Calculations and Explanations (if applicable)

Calculations depend on the explosion-protection concept selected (e.g., Ex d, Ex e, Ex i), which has not yet been confirmed with the ExCB, and on final enclosure geometry from the casing development partner. No calculations have been performed.

Status: Not yet available — pending protection-concept confirmation and final enclosure design.

2.6.f · Temperature Group Calculation (Hottest-Point Temperature Estimation)

No hottest-point temperature calculation or measurement has been performed for this product. As general context: MQ-series metal-oxide gas sensors operate using an internal heating element, a class of sensor commonly associated with published operating temperatures in the approximate 200–400°C range — however, this is a general characteristic of the sensor class, not a measured value for the specific sensor models, drive circuitry, and enclosure configuration used in this product. A worst-case hot-spot measurement is identified as the top-priority action required to substantiate the recommended T4 classification in Section 2.5.

Status: Not yet available.

2.6.g · Usage and Installation Instructions (Draft)

SUB-ITEM	STATUS	REMARKS
a) Safety warnings	Not yet available	No Ex-specific safety warnings have been drafted.
b) Installation requirements (cable entry, torque, grounding, cleaning)	Partially available	A mechanical mounting method exists separately (L-bracket, installed to existing structures without drilling or welding — see Section 2.3.c), but cable-entry method, torque values, grounding requirements, and cleaning requirements have not yet been formally specified.
c) Operating instructions and maintenance requirements	Partially available	A firmware command/operation reference exists (Serial, MQTT, and LoRa command protocol for engineering use), but it covers software operation and commissioning — not Ex-specific maintenance or inspection requirements (frequency, content, precautions).

Status: Partially available — not yet consolidated into an Ex-specific installation and operation manual for ExCB review.

2.6.h · Nameplate Information

Nameplate artwork cannot yet be finalized: it depends on the certificate number, protection marking, temperature class, ambient range, IP rating, and serialization scheme — none of which has been assigned yet.

Status: Not yet available.

2.6.i • Ex Component Certificates

No components in this design currently hold an Ex component certificate. As noted in Section 2.3.b, the processing unit (ESP32-S3-WROOM-1U-N16R8) and the communication module (E22-900MM22S) hold RF/EMC certifications (FCC, TELEC, CE, RoHS) — these are not Ex component certificates and do not satisfy this item.

Status: Not yet available.

3. Sample Information

Model, serial number, and sample status (whether the unit can be powered on and operated), together with any required test fixtures or auxiliary equipment.

3.1 • Model, Serial Number, and Status

No formal sample register or dossier for ExCB submission has been established. Internal engineering and bench testing reference the “GLD V2” board configuration under firmware environment gld_v2; a discrete unit serial-numbering scheme for certification samples has not yet been implemented.

Status: Not yet available. The product is still being built from scratch at prototype/development stage; no finalized, serialized unit yet exists to register as a submission sample. This item becomes actionable once a build reaches a stable, submission-ready configuration.

3.2 • Test Fixtures and Auxiliary Equipment

A draft internal functional test plan identifies the minimum equipment anticipated for bench-level verification (this plan has not yet been executed and is not evidence of lab readiness):

TEST GROUP	MINIMUM EQUIPMENT
Firmware / serial	Engineering commissioning tool or serial terminal; firmware package with recorded version
I2C / ADC / DAC	No additional equipment for protocol-level acknowledgement/readback; multimeter or oscilloscope where physical voltage must be substantiated
Power / watchdog timer	Controlled 24 V supply, an applicable battery source, multimeter, and oscilloscope/logic analyzer
Alarm	The actual alarm load, multimeter/oscilloscope, and hearing protection if an audible buzzer is fitted
LoRa	At least one counterpart Cluster Head/Gateway device with recorded configuration
RS-485 / Modbus	An RS-485/USB-RS485 master with proper termination and the agreed register map

Status: Draft plan only — not yet executed, and not evidence of ExCB/laboratory test readiness.

This is a working document, prepared in stages, drafted in direct reference to the IECEx/ATEX Certification Information Requirements (original English/Mandarin version issued by the certification body). Data sources: the official product technical datasheet (Institute of Technology Bandung, Revision 4.0), internal technical specification documentation, EMC parameter measurement data, and product photography. Fields marked “Pending confirmation” are not yet final and must not be relied upon for procurement or certification purposes without further verification.